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Project 3 — Renewable Energy India 360

Is India's Renewable Energy Capacity Keeping Pace with Its Potential?

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Business Problem

- Which Indian States/UTs are leading in renewable energy capacity, and which states have high potential but low utilization?

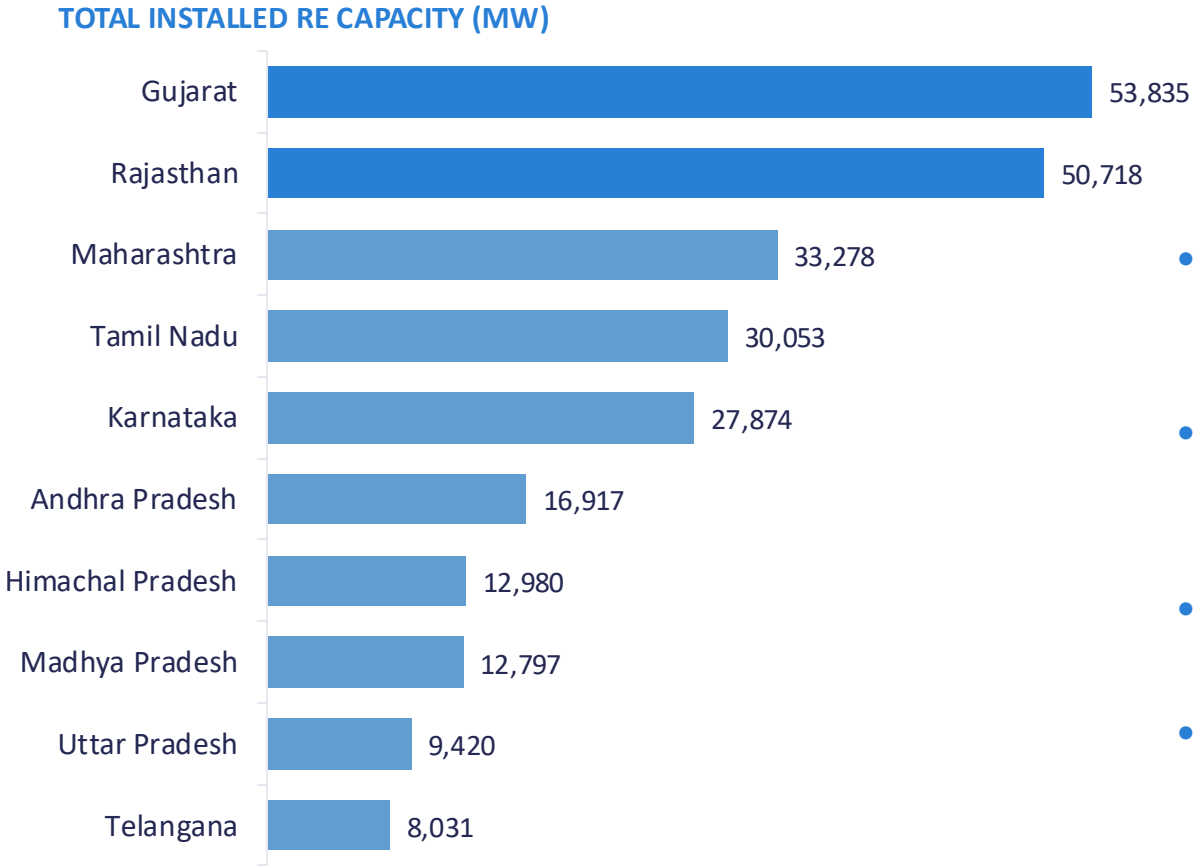
India is rapidly scaling its non-fossil fuel capacity toward ambitious climate targets (500 GW by 2030).

Consulting companies and investors need a data-driven view of regional disparities to optimize capital allocation, grid infrastructure and investments.

Dataset Understanding

- Tracks installed capacity across Solar, Wind, Bio-Power, Small Hydro, and Large Hydro (MW) across 38 states/UTs.
- Tracks estimated resource potential across Wind, Small Hydro, Biomass, Bagasse Cogeneration, Floating Solar, Ground-Mounted Solar, and Large Hydro (MW).

1. Top 10 states: installed capacity at a glance

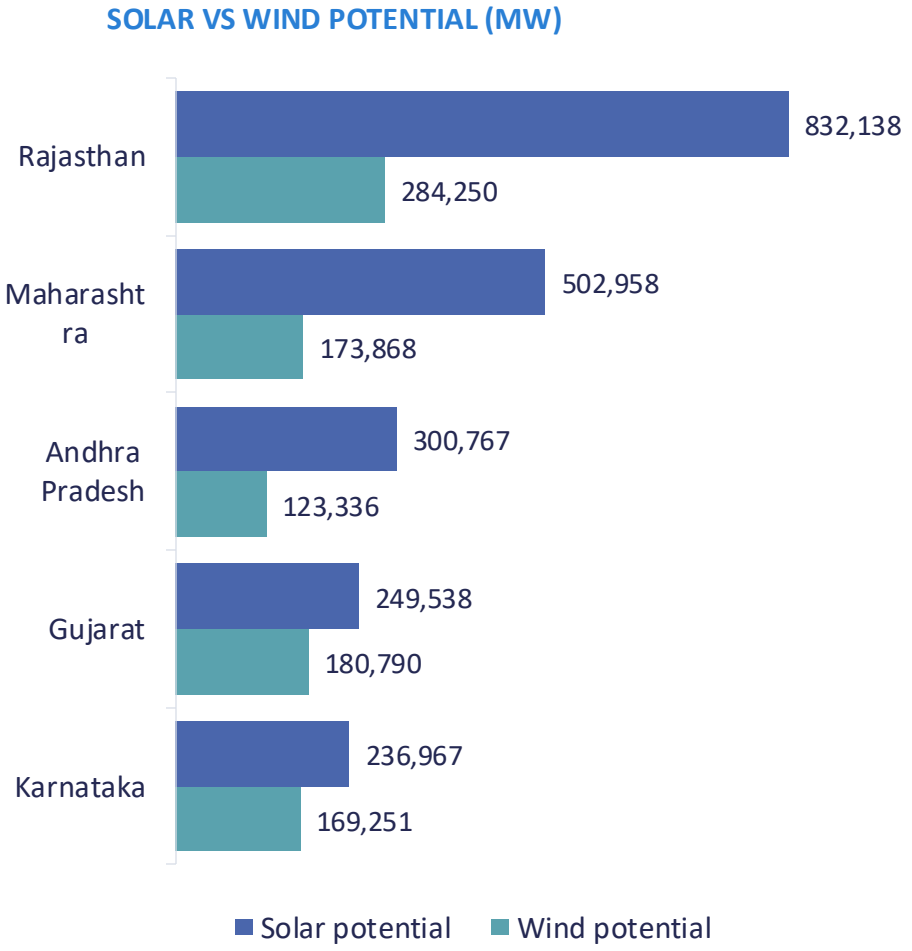


86.58%
of India's capacity in 10 states

35.38%
Gujarat + Rajasthan share

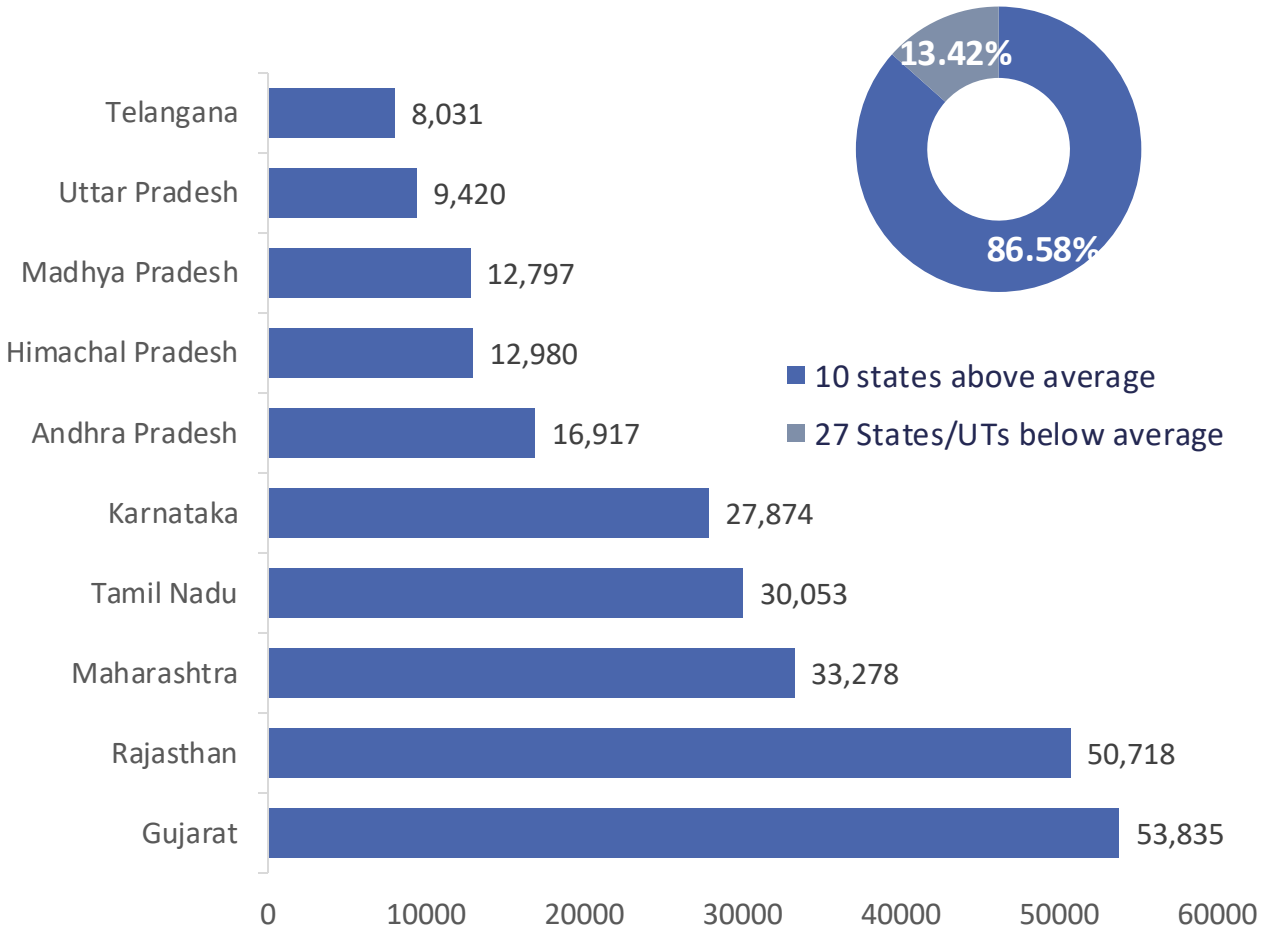
- Gujarat and Rajasthan are India's undisputed renewable powerhouses that is driven by favorable geographical, open land, and strong government backing.
- Large industrial states like Maharashtra, Tamil Nadu, Karnataka are follows closely to meet their massive commercial energy demands.
- The States which have the higher total capacity also have higher production and supply
- These Top 10 states together hold 255,901.59 MW out of India's 295,554.34 MW total capacity—meaning 86.58% of all renewable energy in India is concentrated in just 10 states.
- Alone Gujarat and Rajasthan generate 104,553.17 MW, which is 35.38% of India's entire clean energy capacity.

2. Identify States/UTs where solar potential is higher than wind potential.



- This shows that how much does a State or UG have the Solar potential with compression between Wind potential.
- The State that has highest solar potential and wind potential is Rajasthan and the rest of them are very low value in compression of Rajasthan. So, we can say that Rajasthan have more of a geographical advantage and land so, the Rajasthan have more potential in both solar and wind energy
- The India's solar potential is 3,445,838.10 MW, or 71.7% of total RE potential is nearly 3 times larger than wind potential is 1,163,856.00 MW, or 24.2%.
- Only 3 coastal/island UTs (Andaman & Nicobar Islands, Puducherry, and Lakshadweep) have higher wind potential than solar due to their limited land area.
- There is Wind-Solar Hybrid Opportunity in 5 States: Rajasthan, Gujarat, Maharashtra, Karnataka, and Andhra Pradesh have massive solar potential and massive wind potential
- If the wind potential is very low than the investment will be risky in wind as compared to solar.

3. Find States/UTs having total installed RE capacity higher than the overall average.



- 10 State are above the overall average capacity.
- So that show Gujarat and Rajasthan have more total capacity than any other State
- Means that Gujarat and Rajasthan have more capacity that produce clean energy.
- India's switch to clean energy is uneven. Most of the new green energy is built in just a few leading states, like Gujarat and Rajasthan, while other states are lagging behind due to structural, financial, or grid-connection limitations.
- Only 10 states beat the national average and produce 86.58% of India's green electricity, while the remaining 27 States/UTs have fall below the average and share only 13.42% (39,652.75 MW) of total capacity.

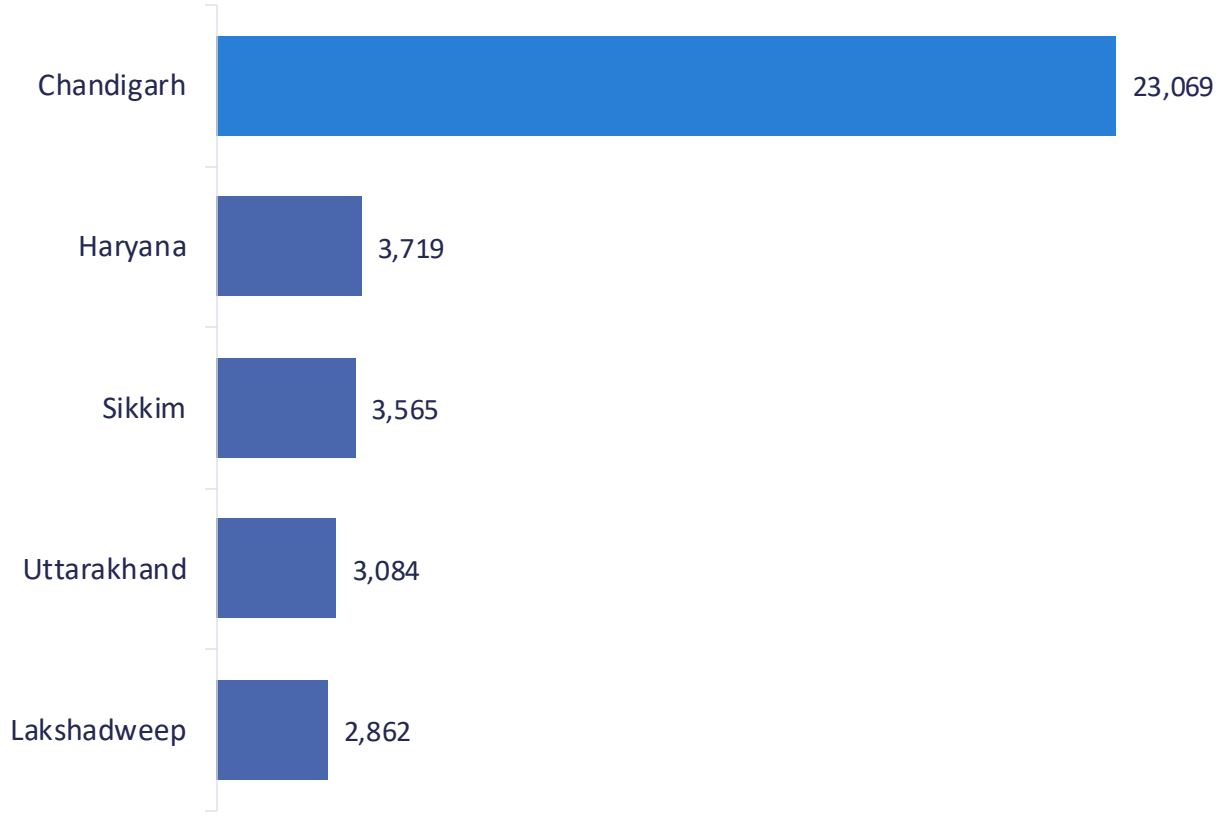
4. Classify States/UTs as High, Moderate, or Low RE Utilization

HIGH (≥ 20%)	MODERATE (≥ 6% to 20%)	LOW (< 6%)
Chandigarh 230.69%	Others 17.34%	Assam 5.97%
Delhi 100.10%	Jammu and Kashmir 15.45%	Maharashtra 4.84%
Haryana 37.19%	Gujarat 12.40%	Rajasthan 4.54%

- High (≥ 20%): States doing better than the average state (19.8%).
- Moderate (≥6% to 20%): States doing better than the national average baseline (6.1%), but below the state average.
- Low (< 6%): States falling below the national average baseline (6.1%).
- Mean (19.8%) = ∑Utilization / Number of States * 100
- National Average (6.1%) = ∑ Total RE Capacity / ∑ Total RE Potential * 100
- Smaller regions like Chandigarh show High utilization because its capacity excides the potential than it has. So, in Chandigarh there is no need for further infrastructure development.
- Target the Low Utilization States for investment because their capacity has not come close to their potential, but they still have massive, untouched room for future expansion.

5. Installed RE Capacity per 10,000 MW Potential (Top 5 States/UTs)

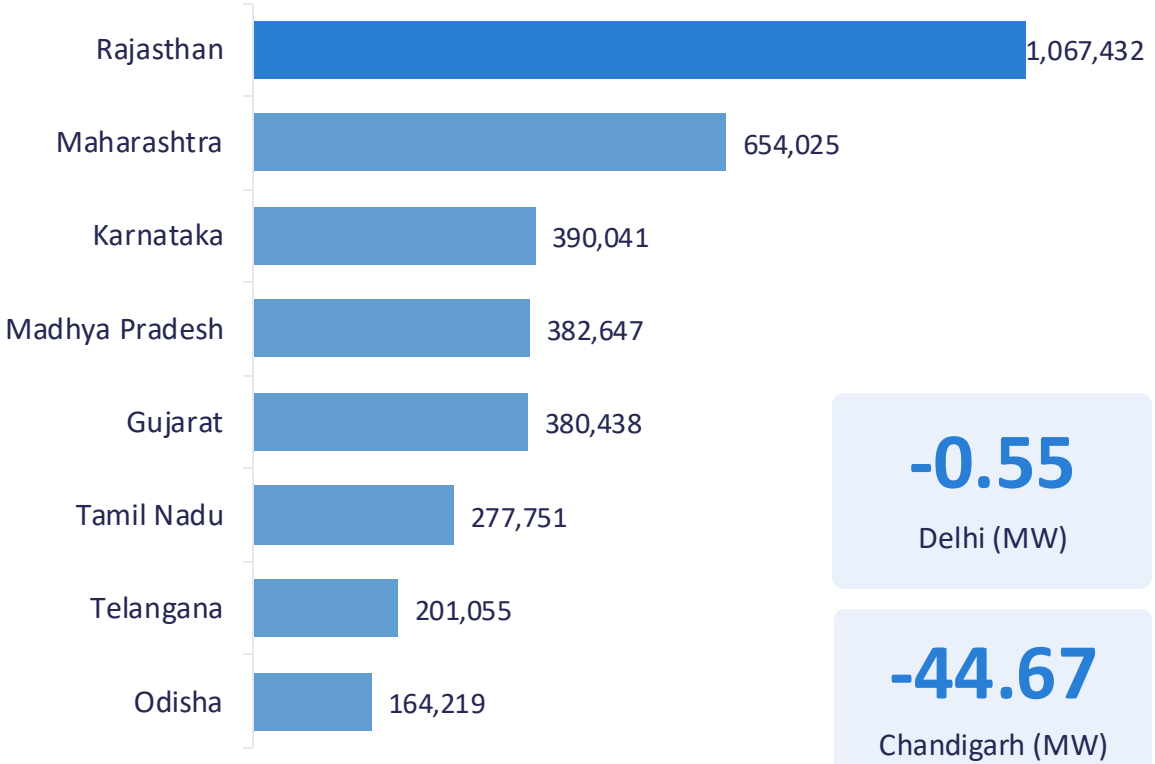
Deployment speed vs room left to grow



- It shows if every state had exactly 10,000 MW of potential, who is building the fastest relative to their size
- So, Chandigarh, Haryana, Sikkim, Uttarakhand and Lakshadweep are deploying infrastructure most aggressively relative to their size.
- But there is still more room for development in Haryana, Sikkim, Uttarakhand and because they have more potential than compared to their capacity. They still have 63% to 69% unused potential for future expansion.

6. Untapped potential and utilization by state

UNTAPPED RE POTENTIAL (MW), STATES IN THE RESULT TABLE



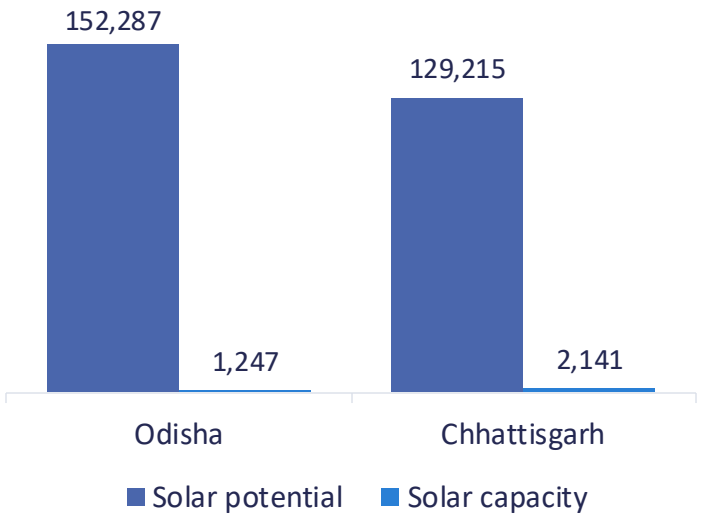
-0.55
Delhi (MW)

-44.67
Chandigarh (MW)

- The Untapped Potential shows how much room for expansion exists in a state before reaching its maximum theoretical clean energy limit. The higher it is the better it to investment in that state as you can see Rajasthan has highest Untapped Potential that indicate that it has 1067431.86 MW of clean energy potential still waiting to be harnessed and developed.
- A higher percentage of Utilization indicates that the state has developed a large portion of its clean energy potential
- Chandigarh (-44.67 MW) and Delhi (-0.55 MW) have negative numbers because they have built more capacity than their baseline land potential.
- So, that leaves on A low percentage such as Rajasthan, Maharashtra, or Andhra Pradesh at under 5% means that this state has massive natural energy resources but only used a tiny fraction of its total potential this shows that this have a major room for future green infrastructure investments.

7. Identify States/UTs with above-average solar potential but below-average installed solar capacity.

Odisha and Chhattisgarh: potential vs built solar capacity



98.34%

Chhattisgarh solar potential

99.18%

Odisha solar potential

- This give the state that have above average solar potential but have below average solar capacity so this give the state that need to increase their solar infrastructure to fully use their potential
- Average Solar Potential= 93,130.76 MW
- Average Solar Capacity = 4,541.62 MW
- Chhattisgarh and Odisha both lie in India's eastern coal belt with cheap thermal power and renewable energy so, it shows that it is good opportunity to invest in solar in this both states
- Odisha has 151,040.12 MW of unbuilt solar potential (99.18% untapped), while Chhattisgarh has 127,073.55 MW of unbuilt solar potential (98.34% untapped).
- Old, empty coal mines have lots of unused, barren land that cannot be used for farming.
- Companies can easily set up large solar plants on this leftover mining land.
- States like Rajasthan and Gujarat are already crowded with solar companies, making land expensive.
- Because very few companies have invested in Chhattisgarh and Odisha so far, new investors can enter early at a lower cost and earn higher profits.

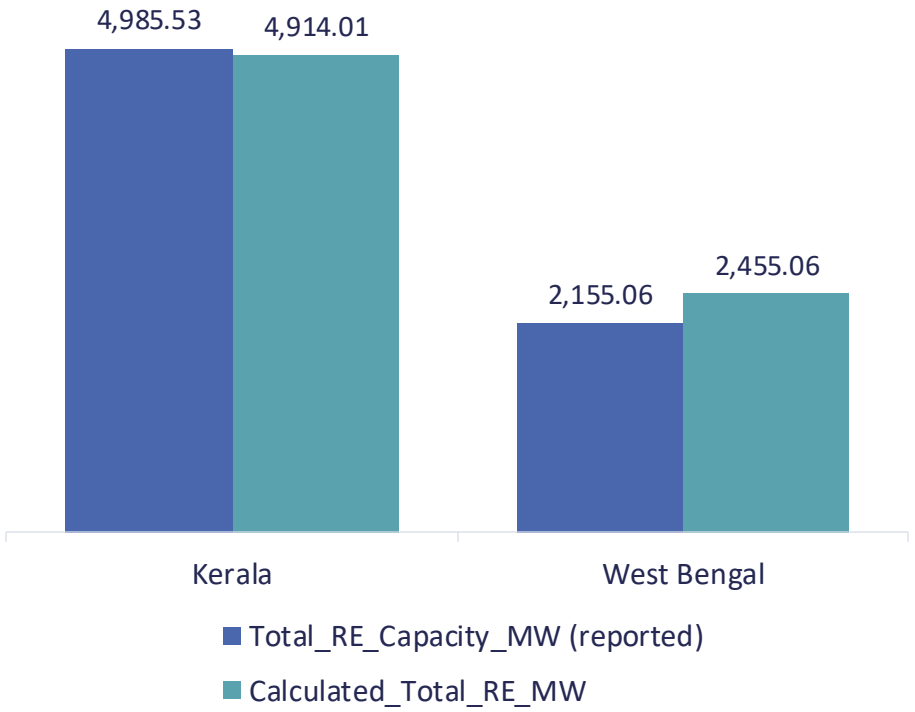
8. Initial Data Exploration: shape, data types and Data Cleaning

INFERENCE

- The `shape` method shows 37 rows and 9 columns in the capacity dataset, and 37 rows and 11 columns in the potential dataset.
- The `dtypes` method checks data types, showing state names as text and energy values as numeric.
- The `isnull()` and `duplicated()` methods check missing values and duplicate records; the displayed results show none in the rows shown.
- The `describe().round(2)` method generates summary statistics such as mean, median, minimum, maximum, and standard deviation.
- The statistics show considerable variation in renewable energy capacity across states/UTs.
- This initial analysis helps assess data quality and prepares the datasets for comparing installed capacity with renewable energy potential.
- The `drop(37)` method removes the row at index 37 (Total) from both datasets.
- The `replace()` method standardizes state/UT names, such as "Orissa" to "Odisha" and "NCT of Delhi" to "Delhi".
- This helps maintain consistency in state names before combining the datasets.

9. Kerala and West Bengal: reported vs calculated total

TOTAL RE CAPACITY (MW)



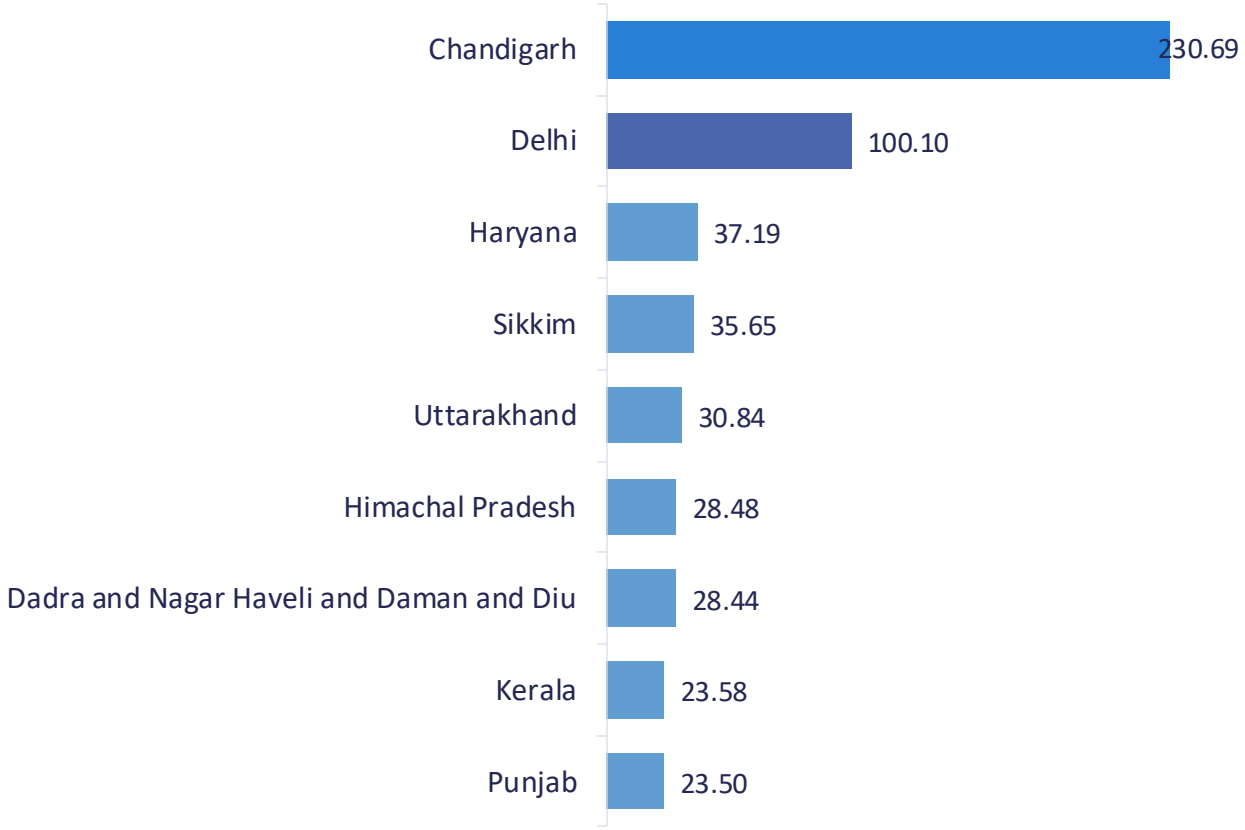
−300.00
West Bengal mismatch (MW)

+71.52
Kerala mismatch (MW)

- The code calculates total renewable capacity by adding solar, wind, bio-power, small hydro, and large hydro capacities.
- It compares the calculated total with the reported total using Capacity_Mismatch_MW.
- Kerala has a difference of 71.52 MW which indicates that its Total_RE_Capacity is more than its Calculate_Total_RE_MW, while West Bengal has a difference of −300 MW which indicate that its Total_RE_Capacity is less than its Calculate_Total_RE_MW
- Inference:. There are data entry or reporting inconsistencies in the primary dataset for Kerala and West Bengal. These discrepancies should be verified against the original data before using the totals for decision-making

10. Utilization summary: mean, median, min and max

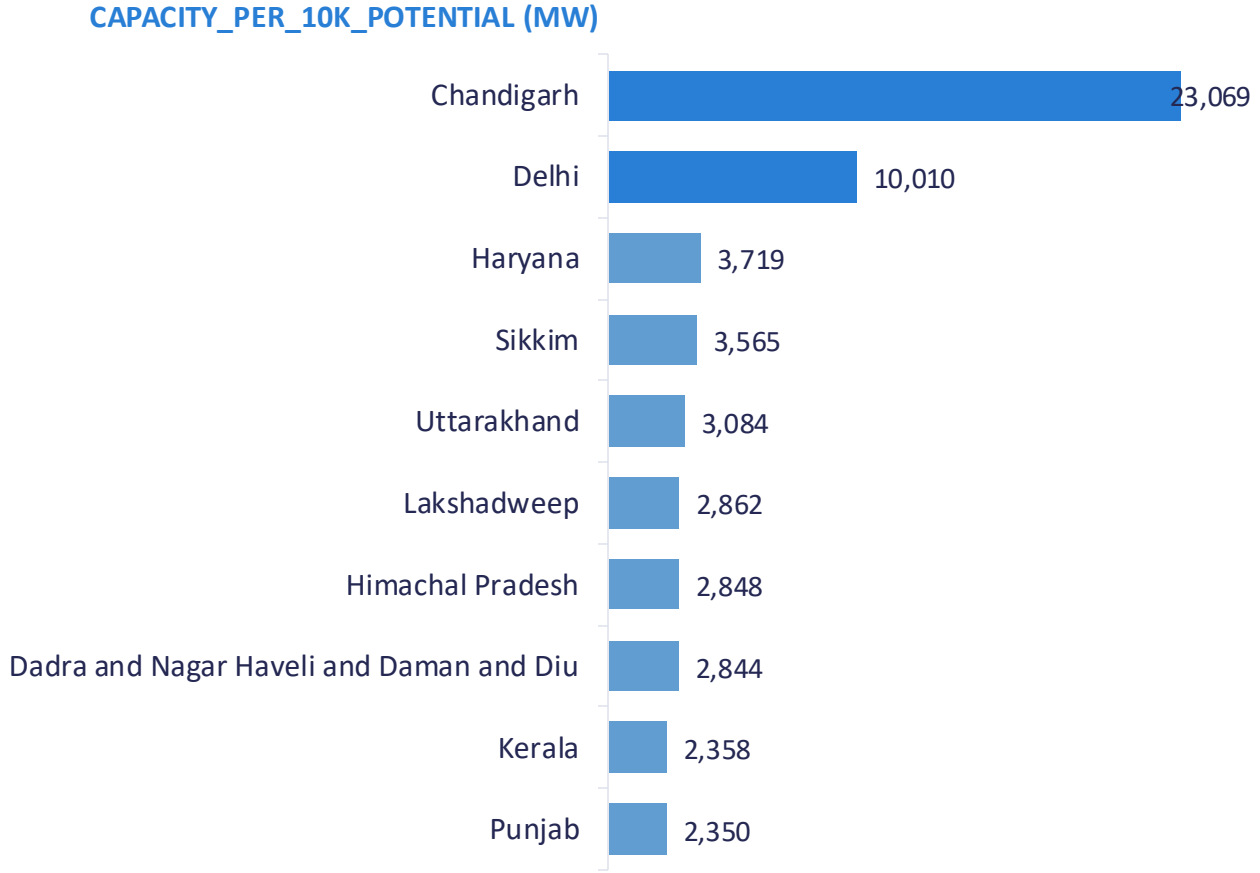
RE_UTILIZATION (%), ROWS SHOWN IN THE OUTPUT



19.7%	6.67%	0.7%	230.69%
mean	median	min	max

- From this states/UTs are shown above the mean utilization; Union territories and smaller regions like Chandigarh (230.69%) and Delhi (100.10%) exhibit exceptionally high utilization rates well above normal bounds
- Inference: Utilization rates exceeding 100% (such as Chandigarh and Delhi) indicate that either installed capacities have surpassed historical potential estimates, or the underlying potential metrics require recalibration due to urban expansion, rooftop solar adoption, or imported green energy credits.

11. Top 10 by capacity per 10,000 MW of potential



- Chandigarh ranks highest at approximately 23,069 MW per 10,000 MW of potential, followed by Delhi at 10,010.
- Inference: Highly urbanized regions or smaller administrative zones utilize a drastically higher proportion of their estimated potential compared to larger states, highlighting distinct geographical constraints and adoption rates. These unusually high ratios should be checked, as they may reflect data inconsistencies rather than actual utilization.

12. The three categories with examples from the output

RENEWABLE LEADER	POTENTIAL GAP	GROWING MARKET
RE_Utilization 37.19% Capacity per 10k: 3,718.77	RE_Utilization 12.40% Capacity per 10k: 1,239.65	RE_Utilization 3.94% Capacity per 10k: 393.97

INFERENCE

- Utilization of 20% or more is classified as "Renewable Leader"; utilization of at least 7% is classified as "Potential Gap"; other records are labelled "Growing Market".
- Inference: This classification helps prioritize regions for further analysis, although the categories depend on the chosen thresholds. The majority of early-stage rows fall under the Growing Market category (low utilization < 7%), representing untapped opportunities for future renewable energy investments and infrastructure scaling.
- At last the final CSV dataset can be reused for reporting, visualization, dashboarding, and further analysis.

Combined Insights

- Renewable capacity is heavily concentrated in Western and Southern corridors (Gujarat, Rajasthan, Tamil Nadu, Karnataka, Maharashtra has more installed Capacity MW).
- States like Rajasthan (1.11M MW total potential), Madhya Pradesh (395k MW), and Maharashtra (687k MW) possess staggering natural resources but currently exhibit low utilization rates (~3% to 4.8%). This represents the primary growth for India's green energy expansion.
- While UTs (Chandigarh, Delhi) and hilly states (Sikkim, Uttarakhand) show high utilization per potential due to limited geographic footprints and hydro efficiency, large mainland states hold untapped multi-gigawatt reserves.

Final Conclusions & Recommendations

- Prioritize investments in Rajasthan, Madhya Pradesh, Odisha, and Chhattisgarh where solar and wind potential dwarf current installations.
- Encourage developers to build integrated Solar-Wind-Storage hybrid projects in high-capacity corridors (such as Rajasthan, Gujarat, Maharashtra, and Karnataka) to maximize land-use efficiency and ensure steady power supply.
- Utilize barren, non-arable land surrounding legacy coal mines in Odisha and Chhattisgarh for massive solar installations, minimizing land acquisition friction and social conflicts.

Thank You